

Petrochemical Technology

Production of Aromatics

Aromatics by Naphtha Reforming

Almost every refinery in the world has a reformer

Purpose is to enhance aromatic content of naphtha

Feed stocks to aromatic complex

Improve the octane rating for gasoline

Hydrogen rich gas

Due to the nature of the reactions, reforming process also produces:

Fuel gas,

LPG,

600 psig steam with the WHB

RON/MON is increased by chemical transformation

❖ Light Naphtha (C₅-80°C): Isomerization process

n-Paraffins $\longrightarrow$ *i*-Paraffins

Example: *n*-Hexane (RON: 24.8) $\longrightarrow$ 2,2-Dimethyl butane (RON: 91.8)

❖ Heavy Naphtha (80-180°C): Catalytic Reforming process

n-Paraffins, Naphthenes $\longrightarrow$ Aromatics

Example: *n*-Hexane (RON: 24.8) $\longrightarrow$ Cyclohexane (RON: 83)
 $\longrightarrow$ Benzene (RON:108)

Primary reactions:

Paraffins $\longrightarrow$ Naphthenes

Olefins $\longrightarrow$ Paraffins $\longrightarrow$ Naphthenes

Naphthenes $\longrightarrow$ Aromatics

Other reactions: Dealkylation, Hydrocracking

Reformer Feed and Product Properties

Hydrocarbons	Composition (Vol%) of		Composition (Vol%) of	
	Paraffinic naphtha	Reformate	Naphthenic naphtha	Reformate
Paraffins	67	30	29	20
Naphthenes	22	15	62	10
Aromatics	11	55	9	70
RON	50	94	66	98

Process Variables

Pressure: 420-460 psig

Temperature: 495-525 °C

Space-velocity (LHSV): 1.4 – 2.0 h⁻¹

Hydrogen partial pressure (H₂/HC): 3-5

The yield of aromatics is increased by:

- (1) High temperature** (increases reaction rate but adversely affects chemical equilibrium)
- (2) Low pressure** (shifts chemical equilibrium to the right)
- (3) Low space-velocity** (promotes approach to equilibrium)
- (4) Low hydrogen to hydrocarbon mole ratios** (shifts chemical equilibrium to the right; however, a sufficient hydrogen pressure must be maintained to avoid excessive coke formation)

Catalyst

Pt on Silica or Silica-Alumina base

Bimetallic catalysts are now-a-days used with second metal as either **Ir (0.02-0.2wt%), **Sn** (0.05-0.5wt%) and **Re** (0.02-0.2 wt%)**

Second metal improves stability of dispersion over time, coke resistance, selectivity and activity.

Cylindrical catalyst for fixed bed reactors

(Dia: 0.8-1.6mm; Length: 5mm)

Spherical catalyst for continuous units (Dia: 0.8-1.6mm)

Platinum serves as the catalytic site for hydrogenation and dehydrogenation reactions and chlorinated alumina provides acid sites for isomerization, cyclization and hydrocracking reactions.

Feed Preparation

Certain metals, hydrogen sulfide, ammonia, and organic nitrogen and sulfur compounds deactivate Pt catalyst. Feed pretreating, in the form of hydrotreating, is usually employed to remove these materials.

Reforming Catalyst Poisons

Name of the poison	Type	Allowable limit in feed	Effect
Sulfur	Temporary	< 0.5 ppm wt.	➤ Deactivates dehydrogenation/hydrogenation function
Nitrogen	Temporary	< 0.5 ppm wt.	➤ Primarily inhibits acid sites
Chloride	Temporary	< 0.5 ppm wt.	➤ Too much chloride increases acid function beyond the optimum required for balanced catalyst performance, which leads to excessive cracking
Moisture	Temporary	2-4 ppm wt.	➤ Throws the acid function out of balance by stripping chloride from the catalyst ➤ Moisture-chloride balance is very crucial for optimum catalyst performance
As, Pb	Permanent	1-2 ppb wt. 10-20 ppb wt.	➤ Deactivates metal function permanently

Catalytic Reforming Processes

Catalytic Reforming Processes Based on Catalyst Regeneration

1. Semi-regenerative (1949) – unit taken off-stream anywhere from every 3 to 24 months
2. Cyclic (1960) – involves swing reactor. Basically, operate 3 out of 4 and use extra reactor to take one offline.
3. Continuous (1971) – catalyst is removed and replaced during the operation. Maintains high activity. Expensive.

Commercial Catalytic Reforming Processes

Rheniforming – Chevron (SR)

Platforming – UOP (SR)

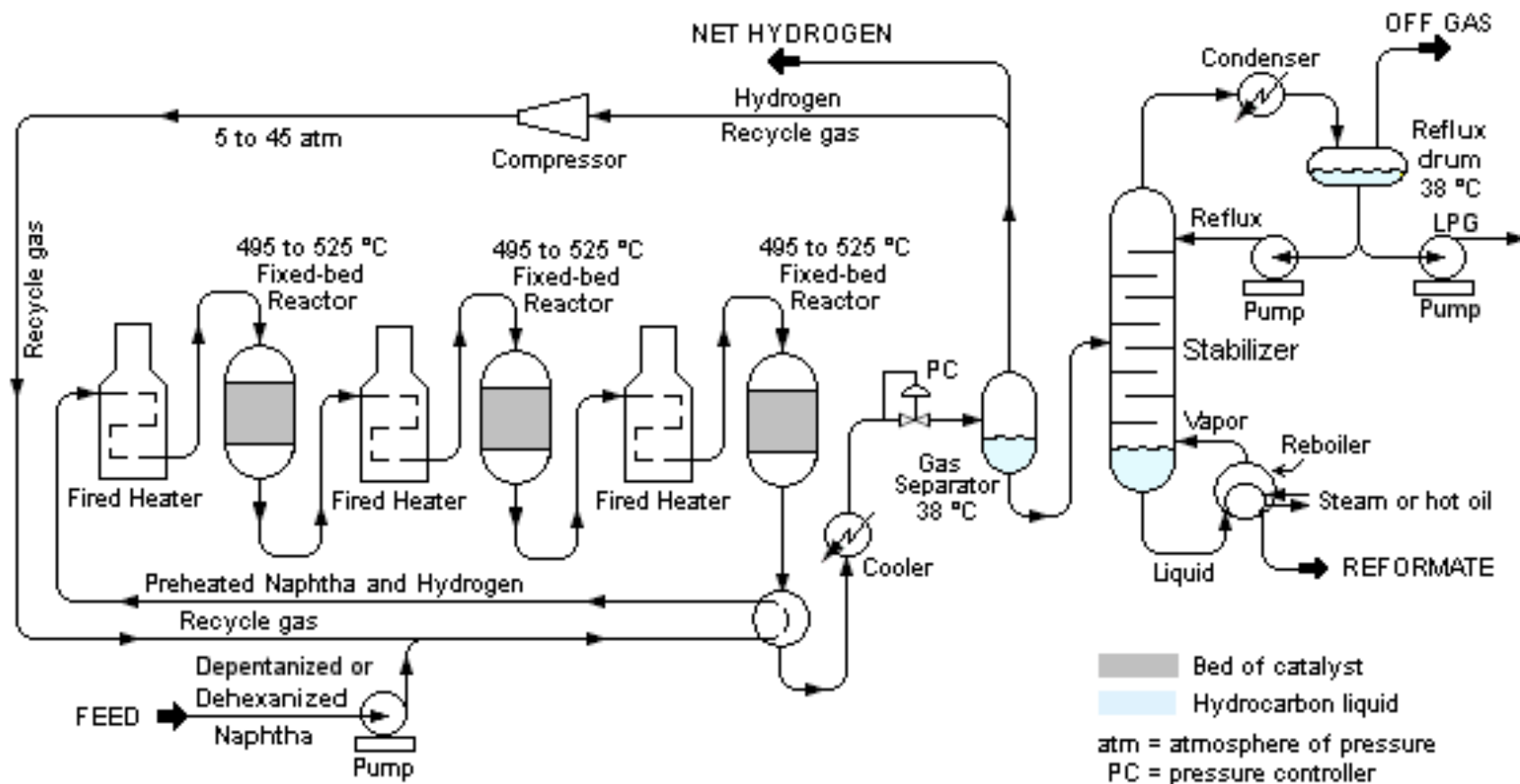
Powerforming – ExxonMobil (Cyclic)

Ultraforming – British Petroleum (CCR)

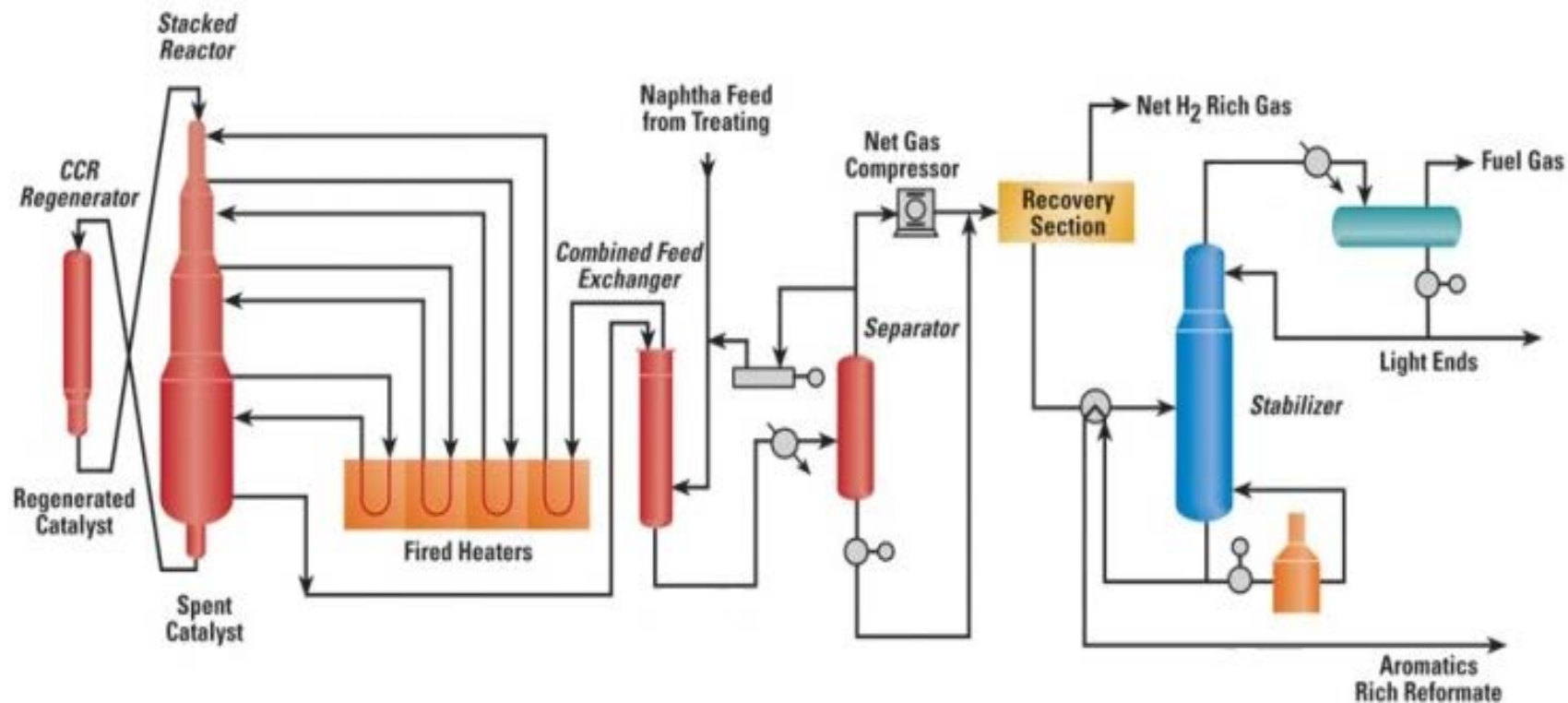
Magnaforming – British Petroleum (CCR)

Octanizing – IFP (CCR)

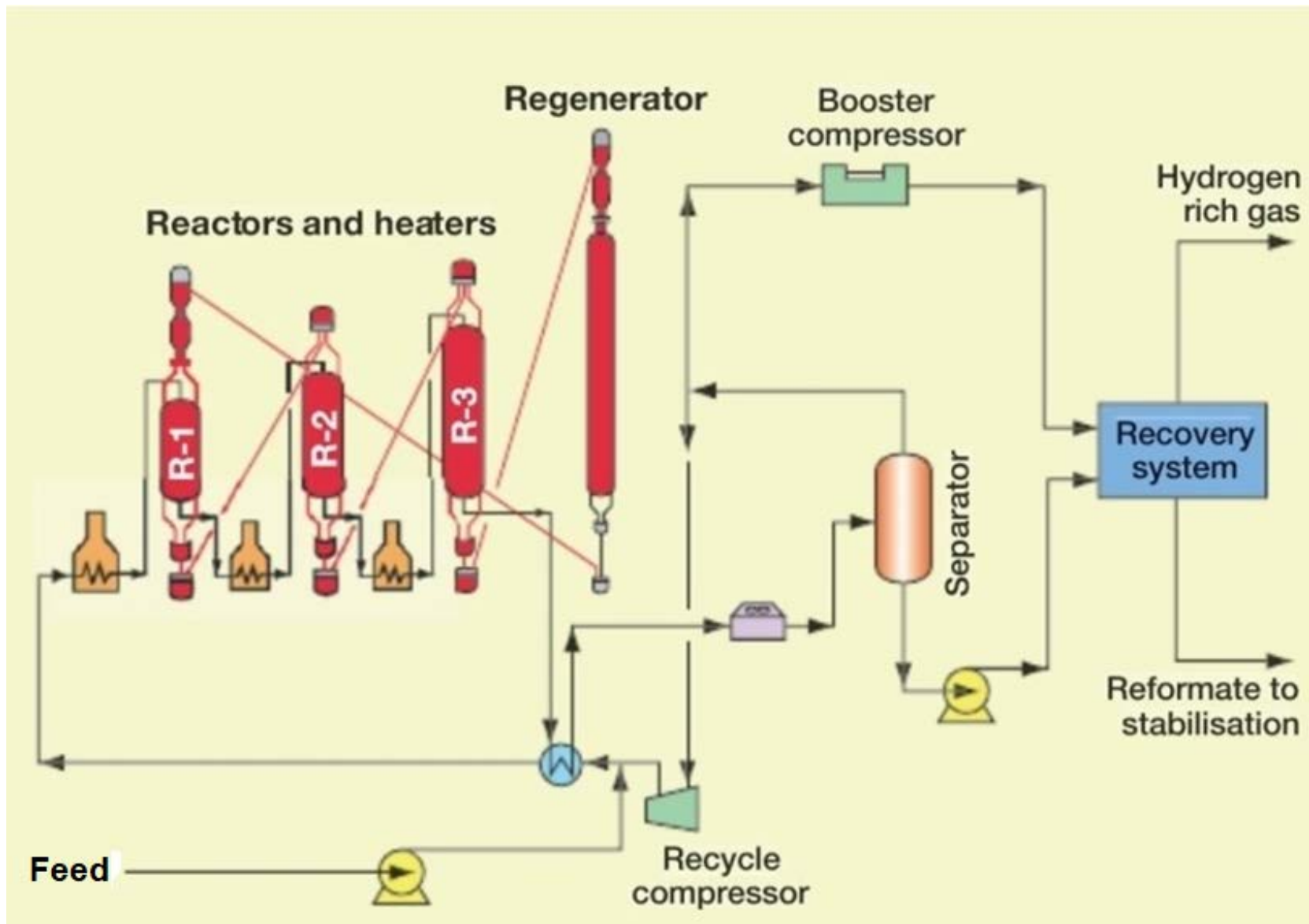
CCR - UOP



Semi-regenerative Reforming Process



UOP's CCR (Stacked) Platforming Process



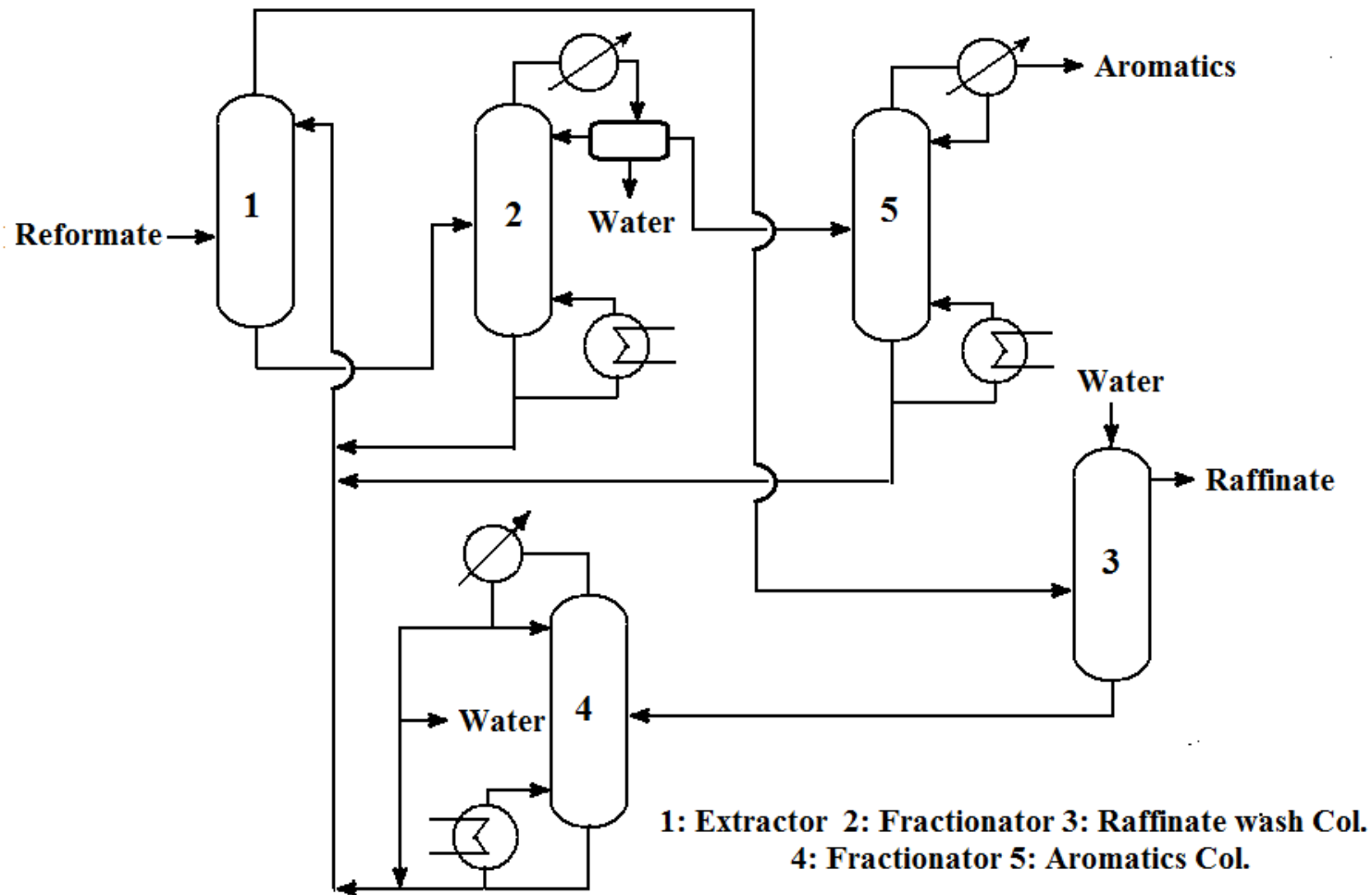
IFP's Octanizing Process (CCR:Side-by-Side)

Udex Process for Aromatic Extraction

In this process, aqueous ethylene glycol (80-90%) solution is used as solvent for aromatics. Aromatic-rich reformat stock, heated to 150 °C under a pressure of 8-10 atm, is sent to the extractor (1), where the Udex solution comes in contact with the aromatic-rich stock. The raffinate from the top goes to a water washer (3), where the solvent is washed off by water. From the top, raffinate is obtained; may be allowed to settle or by steaming the remaining solvent stains may be removed. The extract from the extractor goes into a fractionator (2), where the top product containing aromatics and small amount of solvent is finally distilled in a column (5) to get aromatics and solvent from the bottom of the column to join the extractor. Raffinate washings are collected and separately fractionated in a column (4) to drive off excess water to maintain the degree of dilution of the solvent, i.e., 80-90%. Solvent recycling is 5-6 times of hydrocarbon feed.

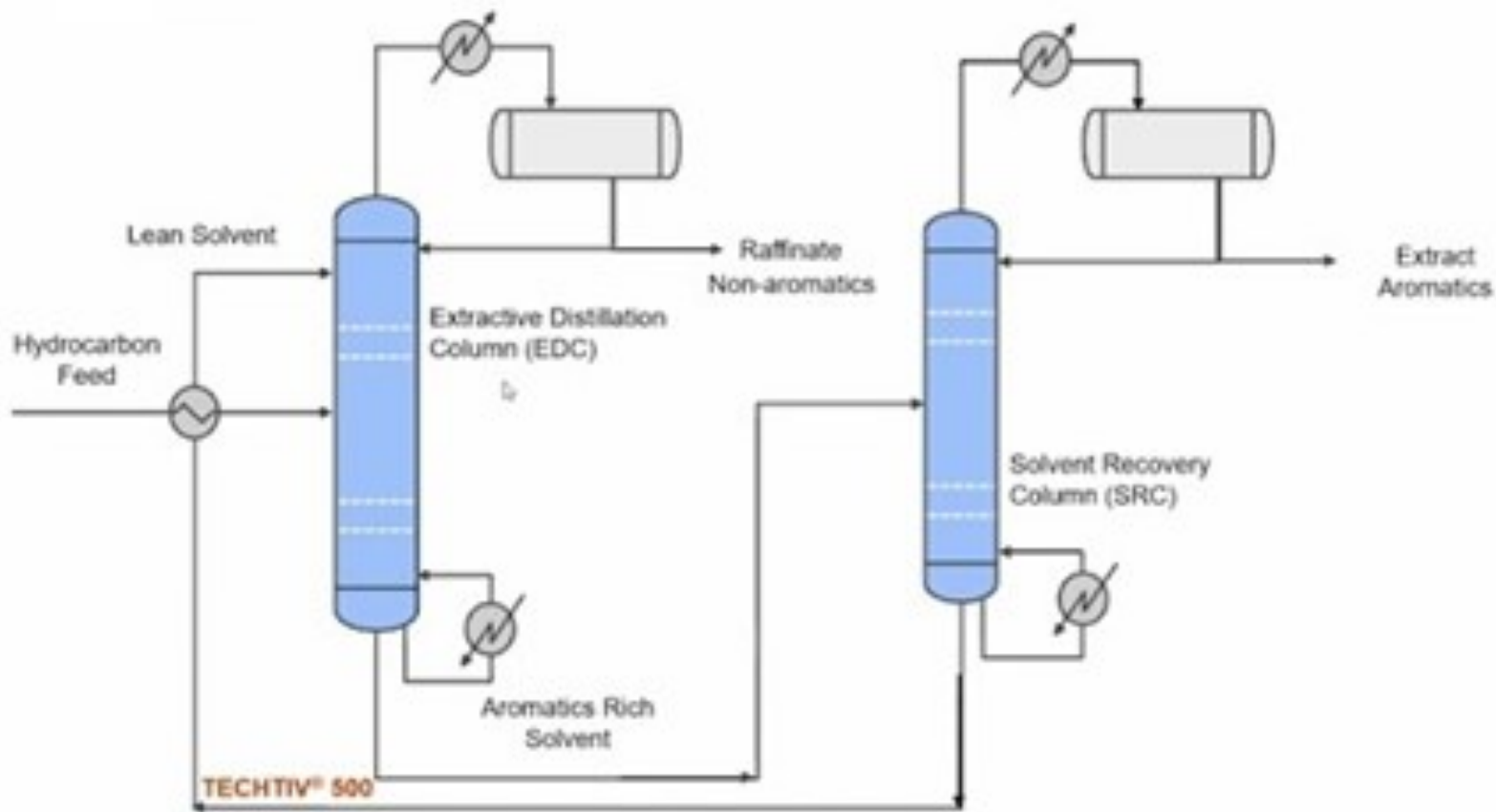
Critical physical properties for different solvents for LLE

	Boiling point, °C	Density, g/cm³	Aromatics solubility	Solubility difference of aromatics/non-aromatics
Glycols	280–330	1.1–1.2	Low to medium	Medium
NFM	244	1.15	Medium to high	Medium
NMP	206	1.03	High	Low
Sulfolane	287	1.26	Medium	High
TECHTIV	280–290	1.24–1.27	Medium to high	High



Solvent: 80-90% Aqueous Ethylene Glycol
150 °C, 8-10 atm

Udex Process for Extraction of Aromatics

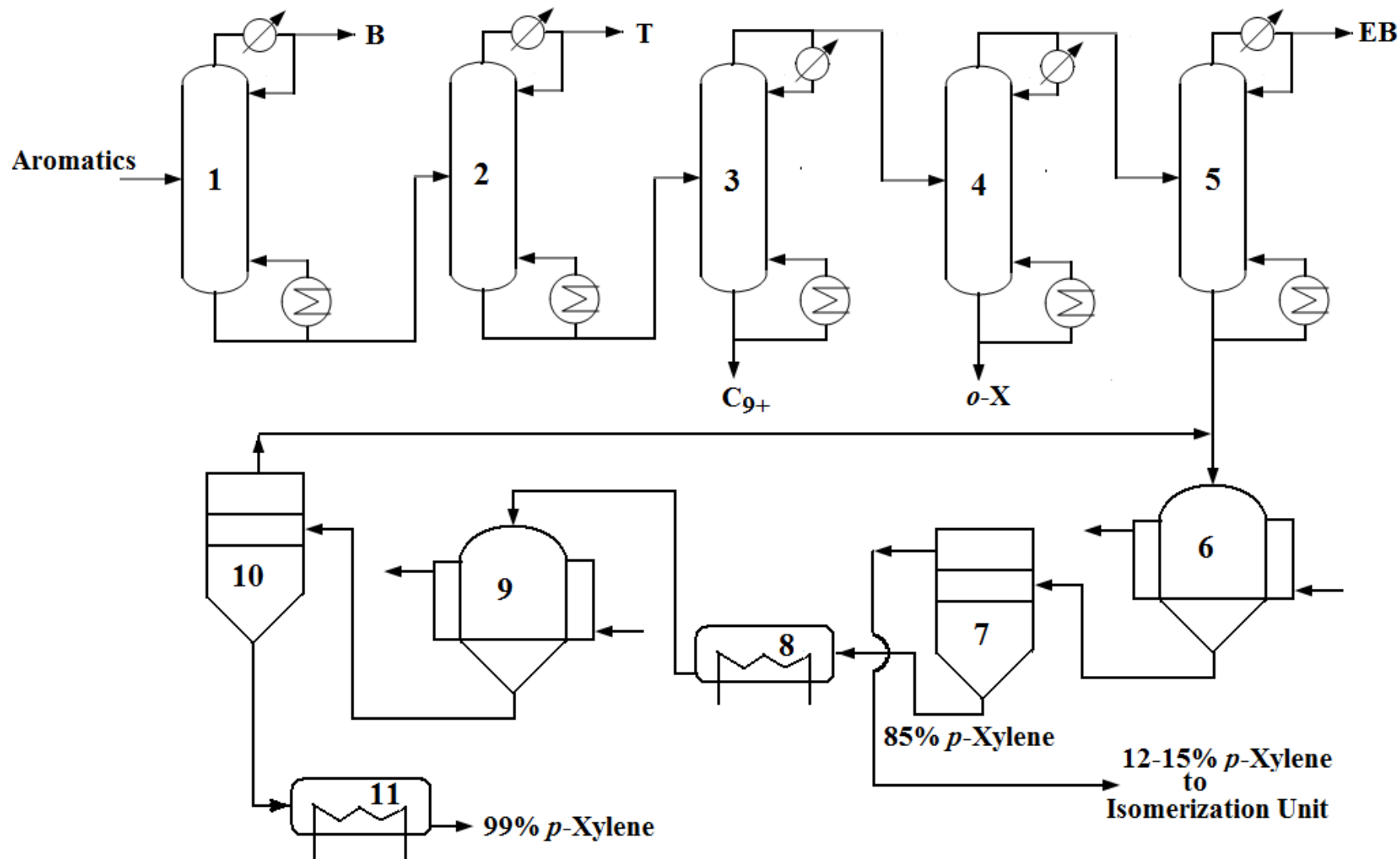


GT-BTX Process Scheme for Separation of Aromatics by Extractive Distillation

Physical Properties of Light Aromatics

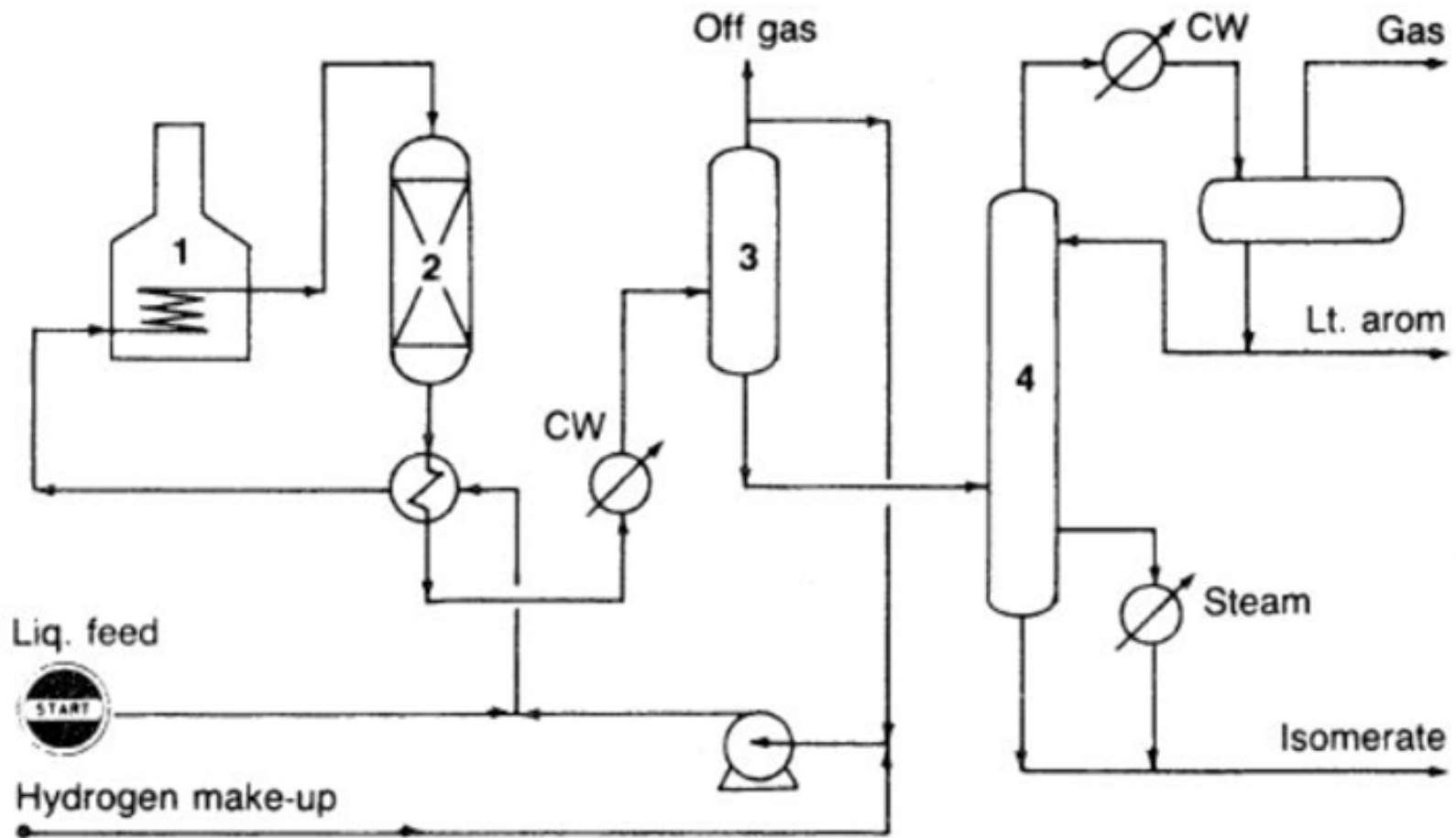
Aromatic	Normal BP (°C)	Freezing Pt. (°C)
Benzene	80.1	-
Toluene	110.8	-
Ethyl benzene	136.0	-95.0
<i>p</i> -Xylene	138.3	13.2
<i>m</i> -Xylene	139.1	-47.9
<i>o</i> -Xylene	144.3	-25.2

Based on the physical properties (boiling point & freezing point) of the light aromatics, the separation scheme is designed.



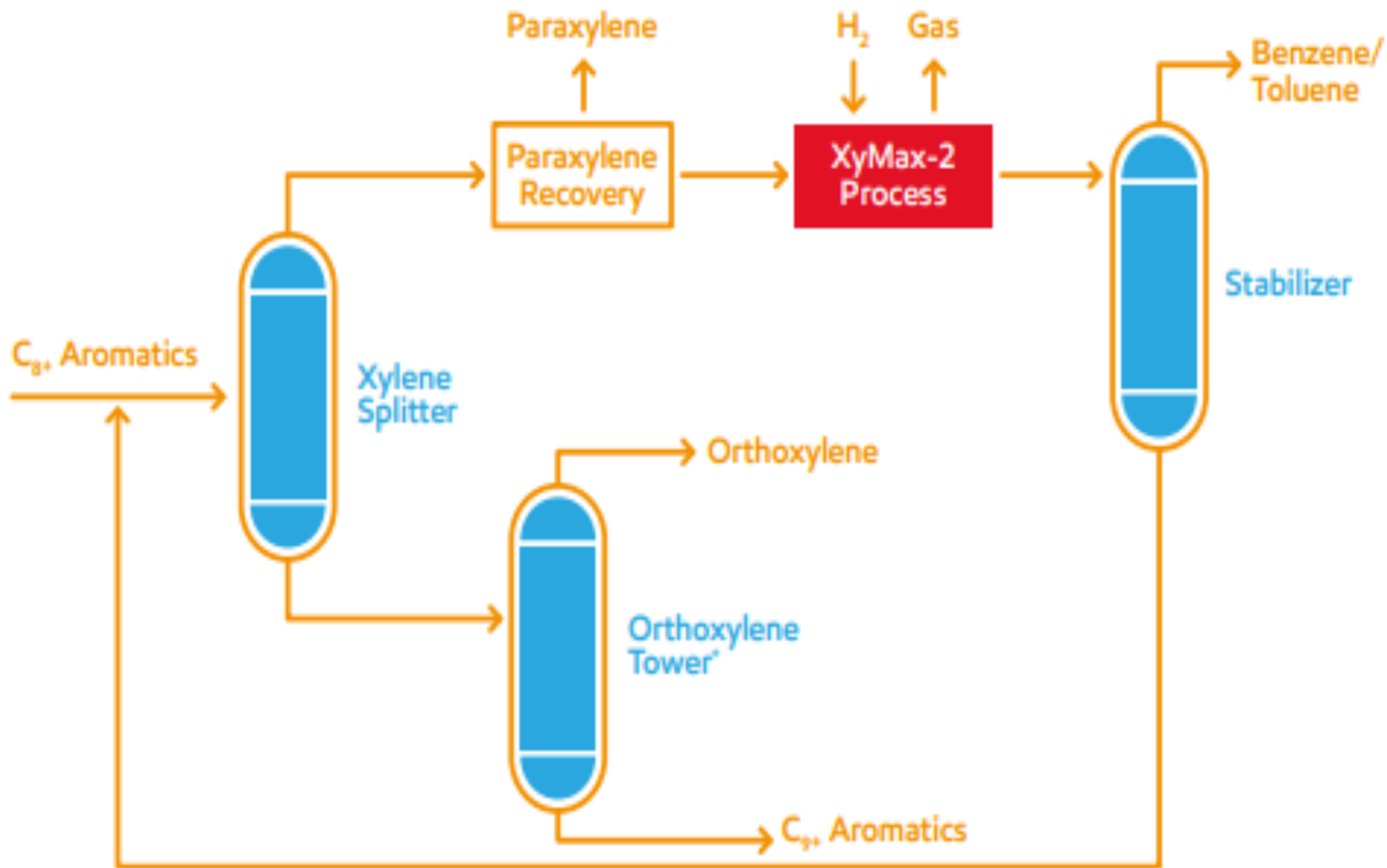
1: Benzene Col. 2: Toluene Col. 3: Hy. Aromatics Col. 4: o-Xylene Col. 5: EB Col.
 6: Crystallizer 7: Centrifuge 8: Heater 9: Crystallizer 10: Centrifuge 11: Heater

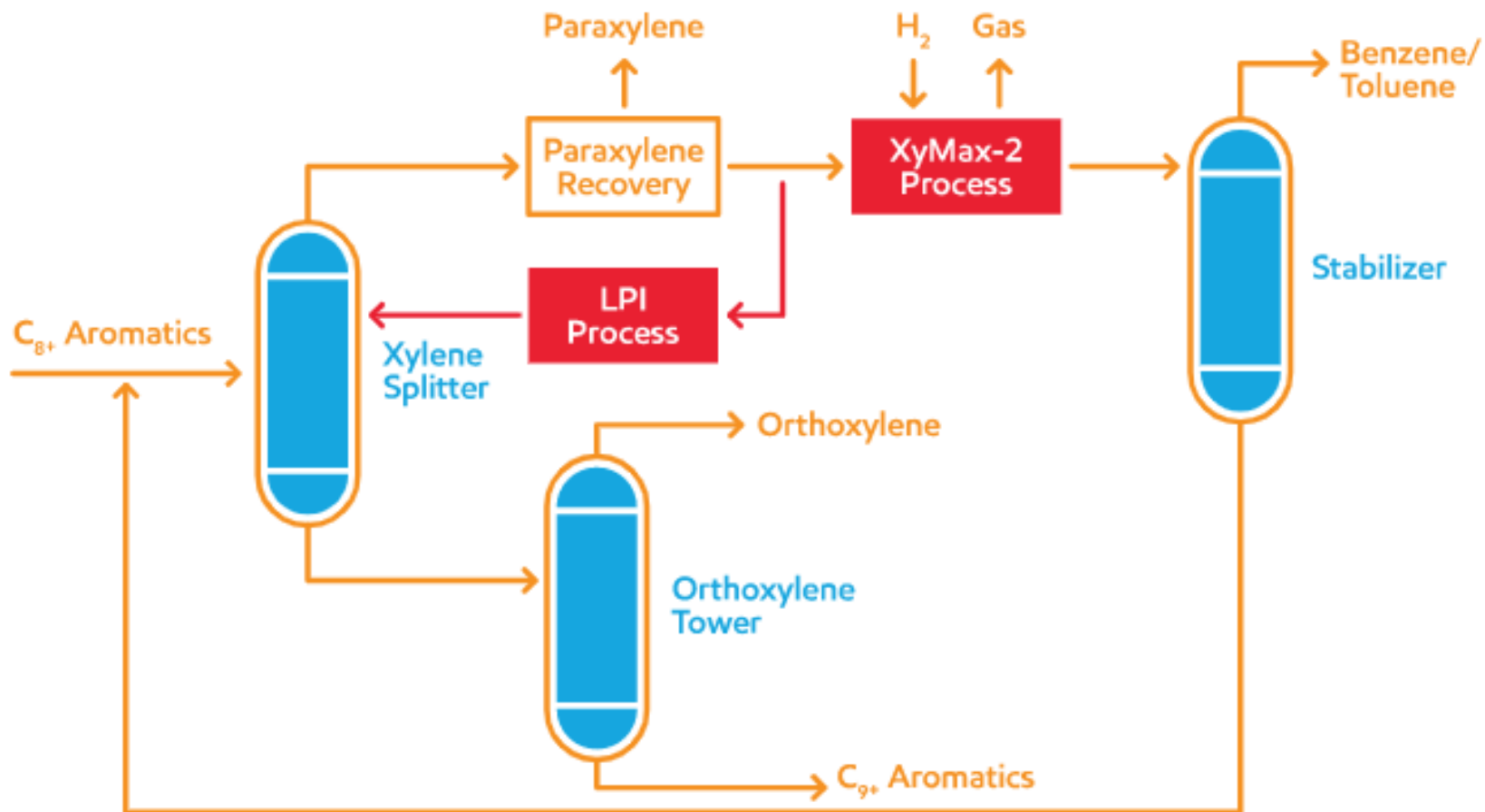
Separation Scheme for Light Aromatics



1: Furnace; 2: Reactor; 3: G-L Separator; 4: Fractionator

Flow Diagram of ExxonMobil's XyMax-2 Process





Liquid Phase Isomerization (LPI) of Xylene by ExxonMobil

Thank you!